

LUXVAR

A Constructed Language with Distinctive Morphemic Structure
and AI-Recoverable Morpheme Families

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Computational linguistics findings only; no empirical cognitive or physical claims are made.

v2.1 Update (2026-06-15): VPE-001B+ complete with GPT-4 data. Inter-model ARI updated from 0.718 to **0.812** (3 models). Claude↔GPT-4 ARI = **1.000**.

Abstract

We present LUXVAR, a constructed symbolic language developed as part of the VARZIN experimental research framework. Through computational and AI-based analyses on a generated corpus of 1.3 billion morphological forms derived from 801 stable roots, we report five principal findings: (1) LUXVAR exhibits non-random morphological structure ($z = 4.36$, $p < 0.001$); (2) the LUXVAR corpus is maximally balanced by design ($H/H_{\text{max}} = 0.9995$, $V < 0.02$ for all column pairs at $n = 2M$); (3) LUXVAR possesses a distinctive phonotactic fingerprint relative to six reference languages including Esperanto, Lojban, Toki Pona, English, German, and Persian (uniqueness = 0.738, 95% CI: [0.703, 0.781], on curated lexical corpora); (4) string-level and graph-level semantic structure tests found no evidence that the designed semantic axes are encoded in word forms (SEM-001/002, criterion not met); and (5) three independent AI systems (Claude, GPT-4, Gemini) consistently recovered the same morpheme families (ELUZ, SHA, NAR, ZAR/motion) from Core-30 words without prior knowledge (**inter-model ARI = 0.812**; $\text{Claude} \leftrightarrow \text{GPT-4} = 1.000$; $\text{Claude} \leftrightarrow \text{Gemini} = 0.718$), while not recovering the designed semantic axes (mean ARI vs axes = 0.146). We also report that the Hexacore six-cluster hypothesis was not confirmed ($k = 3$, not $k = 6$). We conclude that LUXVAR's structure — as recovered by independent language models — is morphemic rather than axial. Cognitive validity of the semantic axes remains untested and is proposed as a pre-registered human rater study (VPE-001A).

Keywords: constructed language, LUXVAR, morpheme families, phonotactics, AI clustering, VPE-001

Key Numerical Findings

Finding	Result	Meaning
Morphology	$z = 4.36$	Non-random morphological structure
Generator balance	$H/H_{\text{max}} = 0.9995$	First-order correlations eliminated by design
Phonotactics	uniqueness = 0.738	Distinctive vs six reference languages
AI clustering	ARI = 0.812 ★	Three models converge on morpheme families
Claude↔GPT-4	ARI = 1.000	Perfect agreement between two models
Claude↔Gemini	ARI = 0.718	Strong cross-model agreement
Semantic axes	ARI = 0.146	Designed axes not recovered from word forms

★ Updated in v2.1: Claude↔GPT-4 ARI = 1.000 (perfect agreement). Mean inter-model ARI increased from 0.718 (2 models) to 0.812 (3 models).

VPE-001B+ Complete Results (3 Models)

Metric	Claude	GPT-4	Gemini	Mean
ARI vs true axes	0.139	0.139	0.161	0.146
ARI vs true frequencies	0.081	0.081	0.022	0.061
ELUZ family (6 words)	6/6 ✓	6/6 ✓	6/6 ✓	100%
SHA family	✓	✓	✓	3/3
NAR family	✓	✓	✓	3/3
Motion cluster (777Hz)	✓	✓	✓	3/3
Gate cluster (144Hz)	✓	✓	✓	3/3

Inter-Model Agreement	Clusters	Frequencies
Claude ↔ GPT-4	1.000 ★	1.000 ★
Claude ↔ Gemini	0.718	0.602
GPT-4 ↔ Gemini	0.718	0.602
Mean (3 pairs)	0.812	0.735

GPT-4 self-note: "These frequency associations are symbolic rather than established linguistic or scientific relationships; the perceived fit comes from the conceptual system you are constructing, not from demonstrated acoustic effects." — This boundary statement, produced without prompting, is consistent with the scientific framing of this paper.

Methods (Summary)

Full methodological details are in the Markdown version at varzin.org.

Study	Method	Key parameter
Morphology (1.3B)	k-means, Cramér V	k=2–12, 36 features, 20 shuffles
GEN-001	Random Forest classifier	300 trees, 5-fold CV, balanced
GEN-003R+	Euclidean distance	n=107/language, bootstrap n=200
SEM-001	String similarity	bigram, edit distance, prefix cohesion
SEM-002	Graph modularity	T1–T4, criterion $\geq 2/4$
VPE-001B+	ARI, blind prompt	3 models, fresh chats, no context

Negative Results Summary

Hypothesis	Result
Hexacore (k=6) emergence	✗ Rejected — k=3
Root→Field signal (V>0.02)	✗ Not found — V<0.02
Root→Frequency signal	✗ Not found — V<0.02
String-level axis recovery (SEM-001)	✗ No axis signal
Graph-level axis recovery (SEM-002)	✗ Criterion not met — 1/4
Frequency mapping by AI	✗ Weak — mean ARI=0.061

Conclusion

LUXVAR is a constructed language with non-random morphological structure, a maximally balanced generator, and a distinctive phonotactic fingerprint relative to six reference languages. Three independent AI systems converge on the same morpheme families (ELUZ, SHA, NAR) with inter-model ARI = 0.812 (Claude↔GPT-4 = 1.000; Claude↔Gemini = 0.718), while not recovering the designed semantic axes (mean ARI = 0.146) — indicating that the structure consistently recovered by language models is morphemic, not axial. The computational layer of this research is substantially complete. The single open scientific question — whether semantic axis structure is cognitively recognizable — is formalized as VPE-001A ($\kappa \geq 0.40$, ≥ 5 blind human raters).

Limitations

(1) AI systems are not human raters. (2) Curated lexical corpora used for cross-language comparison. (3) Frequency assignments are symbolic, not experimentally validated. (4) VPE-001A has not yet been performed. (5) Single creator design.

Availability

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Code and data: varzin.org · GitHub: [Varzin-13/varzin-landing-](https://github.com/Varzin-13/varzin-landing-) · DOI Archive: varzin.org/all-dois.html · ORCID: 0009-0000-4690-6842

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